Universidad Central del Ecuador

Facultad de Filosofía Letras y Ciencias de la Educación

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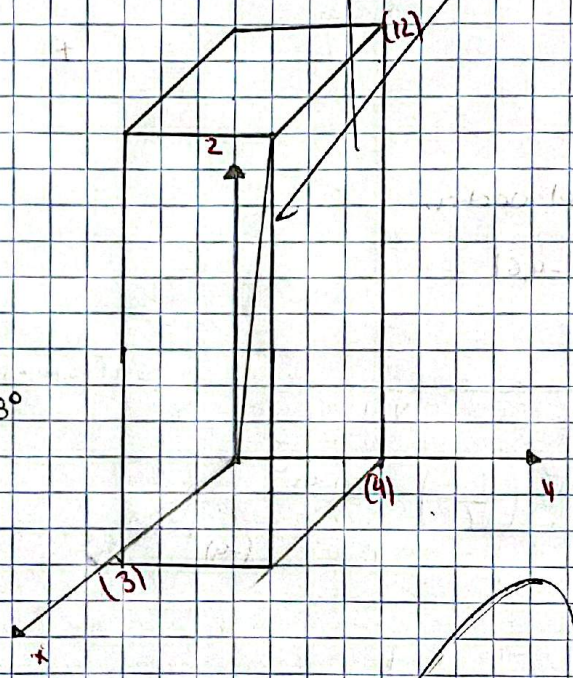
Curso: 5to "B"

Fecha: 22/04/2026

Tema: Vectores en 3 dimensiones

Asignatura: Física Aplicada



① Dado el vector

$\vec{A} = (\overset{x}{3}, \overset{y}{4}, \overset{z}{12})$

Magnitud

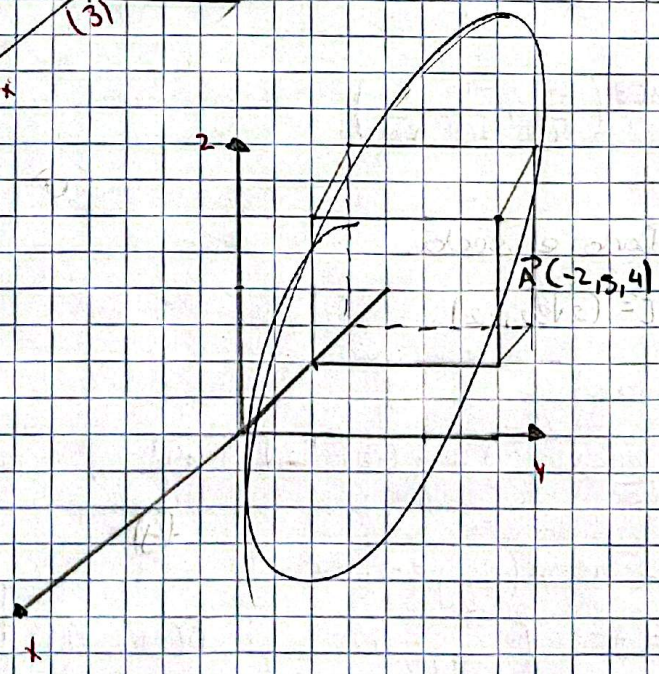
$r = \sqrt{3^2 + 4^2 + 12^2} = \sqrt{9 + 16 + 144}$
$= \sqrt{169} = 13$

$\theta = \arcsin\left(\frac{z}{r}\right) = \arcsin\left(\frac{12}{13}\right) = 67.38°$

$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53.13°$

$\hat{u} = \left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$

$= 0{,}2308^2 + 0{,}3077^2 + 0{,}9231^2$
$= 0{,}04 + 0{,}09$



② Dado el vector

$\vec{B} = (\overset{x}{-2}, \overset{y}{5}, \overset{z}{4})$

Magnitud

$r = \sqrt{(-2)^2 + 5^2 + 4^2} = \sqrt{4 + 25 + 16}$
$= \sqrt{45}$

$\theta = \arcsin\left(\frac{4}{\sqrt{45}}\right) = 36.87°$

$\phi = \tan^{-1}\left(\frac{5}{-2}\right) = 111.80°$

$\hat{u} = \left(\frac{-2}{\sqrt{45}}, \frac{5}{\sqrt{45}}, \frac{4}{\sqrt{45}}\right)$

ESTILO

Escaneado con CamScanner

③ Dado el vector.

$\vec{C} = (\overset{x}{1}, \overset{y}{-1}, \overset{z}{2\sqrt{2}})$

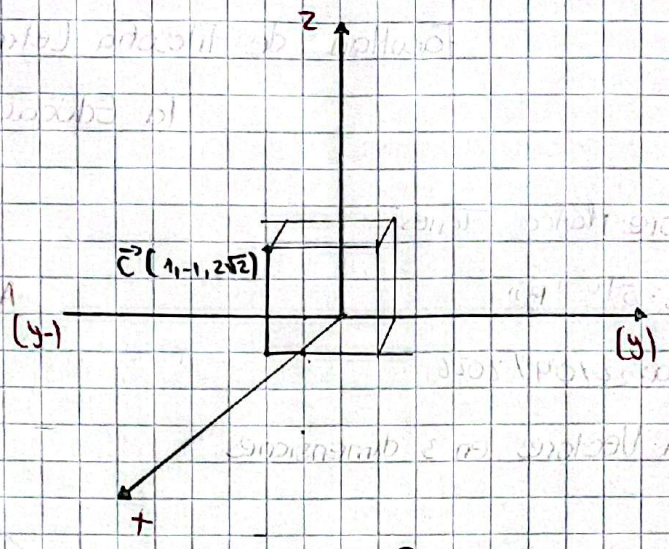
Magnitud

$r = \sqrt{1^2 + (-1)^2 + (2\sqrt{2})^2} =$
$\sqrt{1+1+8} = \sqrt{10}$

$\theta = \arcsin\left(\frac{2\sqrt{2}}{\sqrt{10}}\right) = 63.43°$

$\phi = \tan^{-1}\left(\frac{-1}{1}\right) = 315°$

$\hat{u} = \left(\frac{1}{\sqrt{10}}, \frac{-1}{\sqrt{10}}, \frac{2\sqrt{2}}{\sqrt{10}}\right)$



④ Dado el vector.

$\vec{B} = (\overset{x}{-3}, \overset{y}{-4}, \overset{z}{6})$

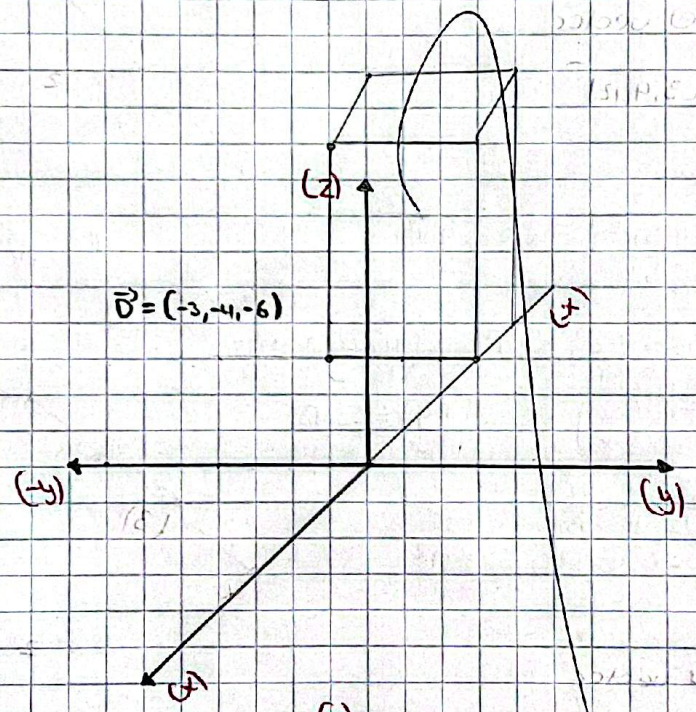
Magnitud

$r = \sqrt{9+16+36} = \sqrt{61}$

$\theta = \arcsin\left(\frac{6}{\sqrt{61}}\right) = 50.42°$

$\phi = \tan^{-1}\left(\frac{-4}{-3}\right) = 233.13°$

$\hat{u} = \left(\frac{-3}{\sqrt{61}}, \frac{-4}{\sqrt{61}}, \frac{6}{\sqrt{61}}\right)$



⑤ Dado el vector.

$\vec{E} = (\overset{x}{2\sqrt{3}}, \overset{y}{2}, \overset{z}{-2})$

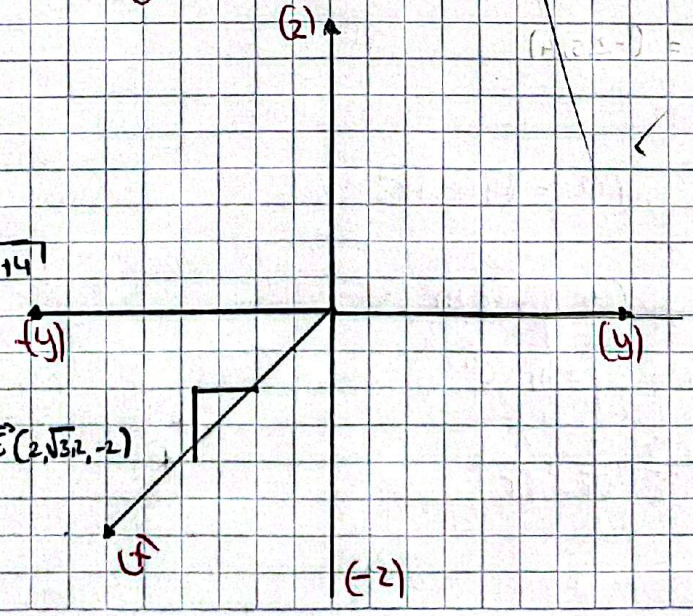
Magnitud

$r = \sqrt{(2\sqrt{3})^2 + 2^2 + (-2)^2} = \sqrt{12+4+4}$
$= \sqrt{20}$

$\theta = \arcsin\left(\frac{-2}{\sqrt{20}}\right) = -26.57°$

$\phi = \tan^{-1}\left(\frac{2}{2\sqrt{3}}\right) = 30°$

$\hat{u} = \left(\frac{2\sqrt{3}}{\sqrt{20}}, \frac{2}{\sqrt{20}}, \frac{-2}{\sqrt{20}}\right)$



Escaneado con CamScanner